Data Communications (CMPU2005)

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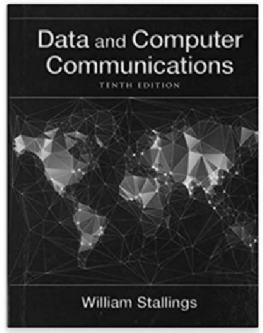
All class material can be accessed via Brightspace.

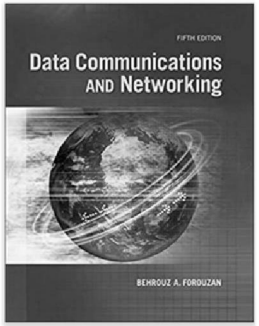
Data Communications

Course Books:

- DATA & COMPUTER COMMUNICATIONS:
  - William Stallings, Prentice Hall.



- DATA COMMUNICATIONS & NETWORKING:
  - Behrouz A. Forouzan, McGraw Hill.



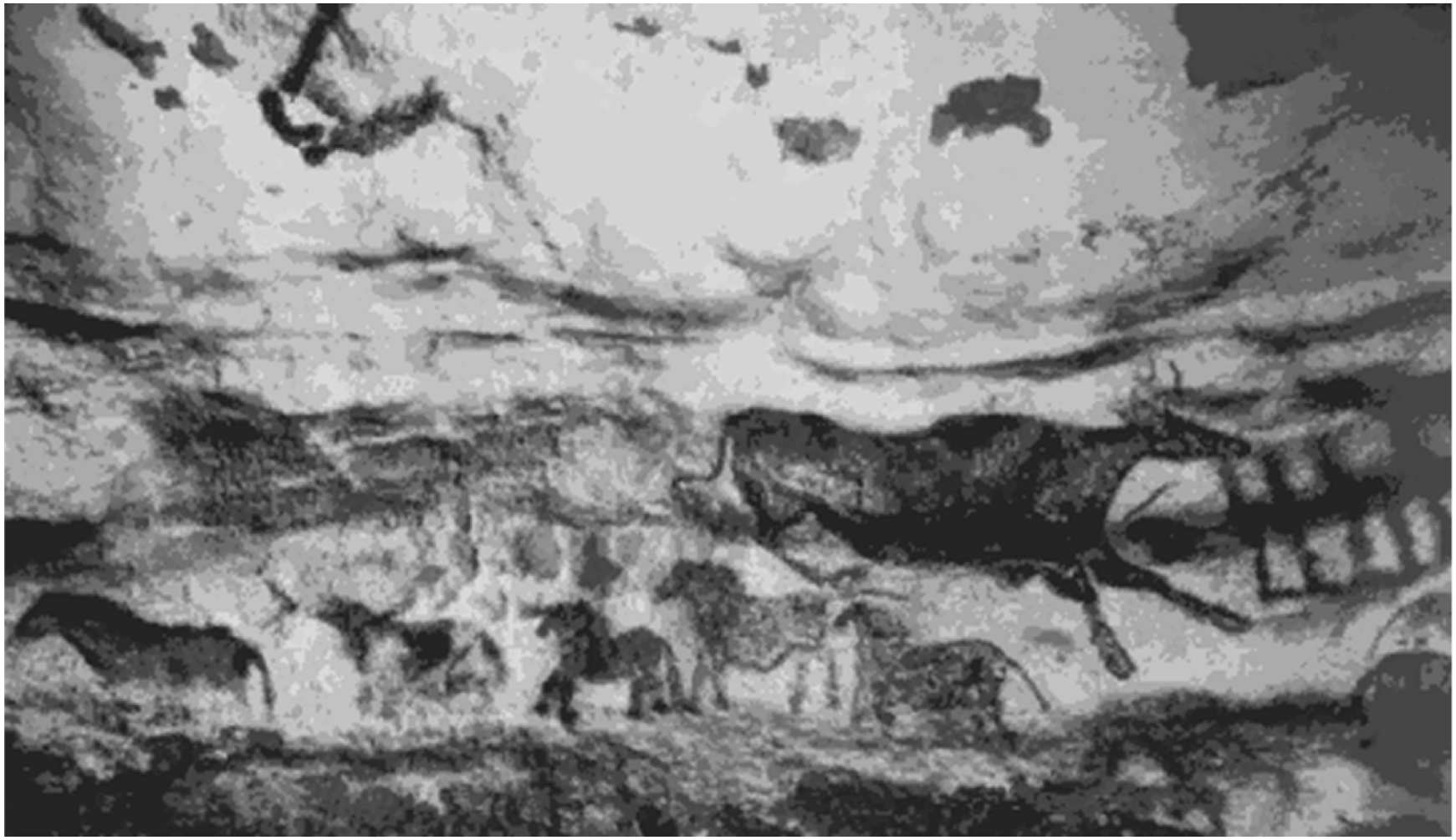
- COMPUTER NETWORKS:
  - Andrew Tanenbaum, Pearson Education.

Introduction

- This course is concerned with the ***Communications Problem***:

"How can two remote entities communicate with each other <u>effectively</u> and <u>efficiently</u>"

Cave Painting - 40,000 to 10,000 BC

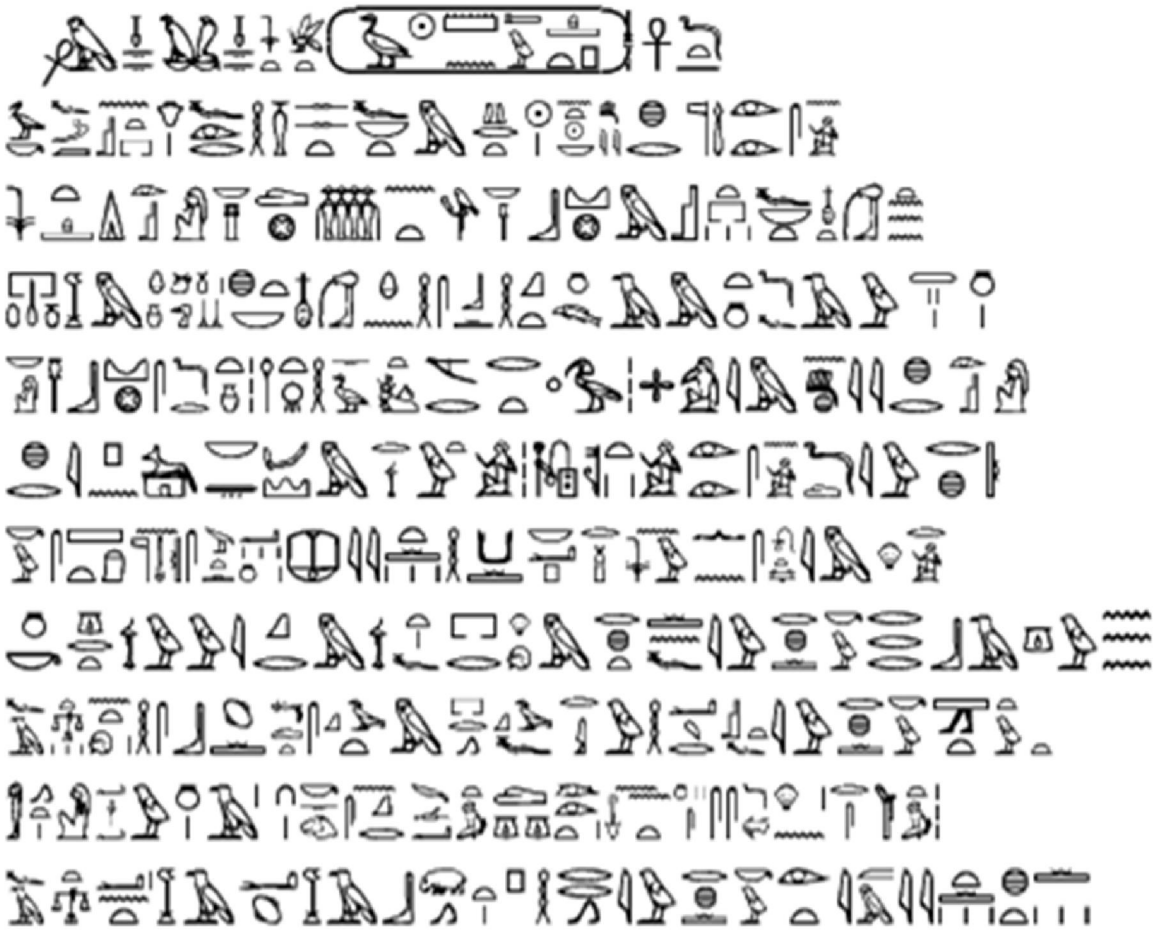


Lascaux, France

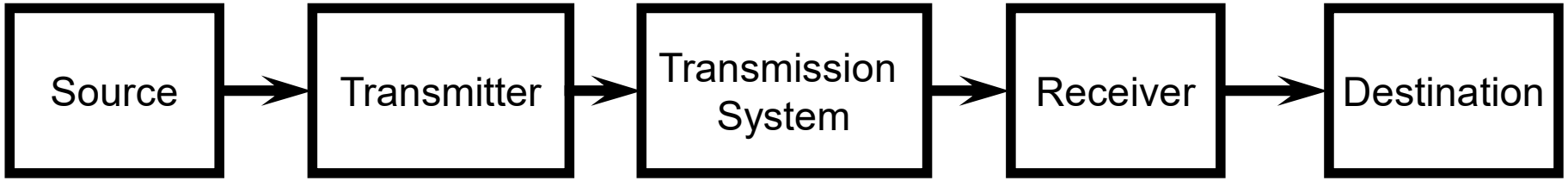
Sumerian Cuneiform - 3000 BC

Inscriptions from the Behistun Rock (Western Iran)

Hieroglyphic Text



Model of Communications System



Communications Model Components

- Source: Device that generates data
- Transmitter: Transforms and encodes data
- Transmission System: The physical system connecting the source and destination devices
- Receiver: Performs reverse function of Transmitter
- Destination: Receives the incoming data

Course Topics

- Introduction
  - » Communications Model/Tasks
- Signal Analysis
  - » Signalling concepts
  - » Bandwidth concepts
  - » Relationship between Data Rate and Bandwidth
- Data Transmission concepts
  - » Digital versus analogue data/signals/systems
- Transmission Impairments
  - » Attenuation/Distortion/Noise
- Channel capacity
- Transmission Media
  - » Wired/Wireless
- Data Encoding
  - » Digital/Analogue Data onto Digital/Analogue signals etc.
- Synchronous /Asynchronous

Course Topics

- Flow Control techniques
- Error detection techniques
- Error Control techniques
- Sample Link Protocol
- Multiplexing
- Switching Networks
- Local Area Networks
  - » Topologies and operation
  - » Protocols
  - » Access Control techniques

……..and much more